Improved Practical Fixed-Time Adaptive Control for Uncertain Robotic Systems With Intermittent Constraint Conditions

Xinjun Wang, *Member, IEEE*, Tengying Yan, Jihang Sui, and Yingying Liu, *Member, IEEE*

***Abstract*—This paper develops an improved practical fixed-time control scheme for uncertain robotic systems subject to intermittent constraints. Existing studies commonly require prior knowledge of the next constraint activation instant upon entering an unconstrained interval. However, in practical robotic applications, the exact instant is often unavailable until shortly before its occurrence. To overcome this limitation, the concept of transition intervals is introduced. Based on these transition intervals, a novel stretch function is designed and integrated with a nonlinear mapping function to effectively address intermittent constraints. To handle the infinite-gain issue in existing prescribed-time stability theorems, a novel time-varying function is introduced, leading to a new stability theorem with prescribed upper bound on the settling-time. Command filtering technologies are employed to mitigate the computational complexity explosion issue in the differentiation of the virtual controller caused by the stretch function, and a filtering error compensation system with prescribed upper bound on the settling-time is developed. The effectiveness of the proposed control scheme is demonstrated through an example involving a two-link rigid robotic system.**

***Note to Practitioners*—Many robotic tasks involve intermittent constraints. For instance, a manipulator performing pick-and-place operations operates between an open workspace and a confined assembly region; a welding robot alternates between free-motion segments and constrained welding paths. In these applications, the tracking error must satisfy strict bounds during constrained phases and be relaxed during unconstrained phases. Existing approaches commonly require prior knowledge of the next constraint activation instant upon entering an unconstrained interval. To address this issue, a stretch function based on the transition interval is introduced, reducing the conservativeness associated with a priori switching-time information and improving applicability. In addition, an adaptive mechanism compensates for external disturbances and modeling uncertainties. Simulation results on a two-link robotic manipulator verify the effectiveness and practicality of the proposed approach.**

***Index Terms*—Adaptive control, intermittent constraints, robotic system, practical fixed-time control, time-varying gain function.**

I. Introduction

In RECENT years, robotics has advanced rapidly, with significant progress in both robotic structures [1], [2] and robotic materials [3], [4]. Along with this progress, application scenarios have continued to expand [5], and the demands on robot control algorithms have increased accordingly. Robotic systems are typically modeled as highly coupled multi-input multi-output nonlinear systems, for which control algorithm design usually needs to account for system uncertainties and unknown external disturbances [6]. Despite these challenges, considerable progress has been made in robotic control, with approaches including PID- [7], sliding mode- [8], intelligent- [9] and adaptive control [10]. However, research on intermittent constraint control for robotic manipulators remains relatively limited. In tasks such as welding and object handling, high tracking accuracy is typically required during welding, grasping, and placement, whereas the accuracy requirements can be relaxed during transfer motions between task locations. Therefore, strict tracking-error bounds need only be imposed during specific task phases rather than throughout the entire operation. Compared with continuous tracking constraints, designing control algorithms under intermittent constraints is more challenging. Research on intermittent constraint control remains limited. Reference [11] conducted the first study on this issue by proposing a piecewise continuous stretch function and incorporating it into a barrier function, thereby enabling tracking control of multi-input multi-output nonlinear systems with a single constrained time interval. Subsequently, [12] designed an improved stretch function, making the resulting control scheme applicable to tracking tasks involving arbitrary constrained intervals, while [13] and [14] further developed the stretch function into a more concise form. In robotic systems, research on intermittent constraint control has also yielded several results. Reference

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[15] applied the stretch function proposed in [11] to the adaptive tracking control of flexible robotic arm systems. Reference [16] proposed a new stretch function to address tracking problems involving a single constrained time interval and incorporated the proposed function into an optimal control scheme for robotic arm systems. Nevertheless, these methods rely on a restrictive assumption that the exact timing of the next constraint activation is available when entering an unconstrained interval, which is often impractical in complex dynamic environments.

Over the past decade, the convergence speed issue of robotic systems has garnered substantial attention, leading to the development of numerous finite-, fixed-, and prescribed-time (PT) control schemes [17], [18], [19], [20], [21]. In [19], a finite-time control scheme was developed for robotic manipulators, achieving rapid system convergence. However, the settling time of finite-time control depends on the system's initial conditions, such that larger initial states may result in a longer convergence duration [22]. To overcome this limitation, a fixed-time control scheme was designed for robotic manipulators in [20], which eliminates the dependence of the settling time on the initial conditions. Nevertheless, the upper bound of the settling time remains determined by the control parameters and thus cannot be arbitrarily specified. In contrast, the PT control scheme has emerged as a significant advancement in modern control theory, owing to its distinctive capability to arbitrarily predetermine settling-time upper bound independent of both system initial states and control parameters. The concept of PT stability was initially introduced in [23] for nonlinear systems in "normal form" through state scaling using a time-dependent function. This pioneering work has stimulated extensive research into PT control methodologies [24], [25], [26], [27]. For example, [26] developed a prescribed-time adaptive neural tracking controller for uncertain manipulators with dead-zone inputs. In [27], a time-varying gain function is incorporated into both the actor–critic learning framework and the controller to achieve learning and state convergence within separately prescribed times. However, the time-varying functions employed in the aforementioned studies tend to infinity as time approaches the PT. This feature makes the system sensitive to measurement noise near the PT [28]. Prescribed-performance-based approaches provide an alternative for achieving a specified tracking accuracy within a PT without employing such singular time-varying gains. [29] uses a piecewise prescribed performance function to drive the tracking errors into a specified region within the prescribed time. Recently, [30] introduced a bounded time-varying gain into a fixed-time control framework, allowing the settling-time upper bound to be specified while avoiding infinite gains. However, the control scheme presented in [30] fails to address system uncertainties.

Traditional time-triggered control strategies generate frequent communication between controllers and actuators. To reduce unnecessary communication resource consumption and extend equipment service life, [31] proposed an event-triggered control scheme. Subsequently, this mechanism has been incorporated into numerous control frameworks for nonlinear systems [32], [33]. The design of a practical fixed-time control scheme with prescribed upper bound on the settling-time (PUBST) for robotic systems subject to intermittent constraints under the event-triggered mechanism would significantly enhance the practical applicability of the proposed controller. This paper employ command filtering technology, which was first proposed in [34] and has since been widely applied in various control schemes [35], [36], [37].

To address the aforementioned challenges, this paper develops a novel event-triggered practical fixed-time control scheme with PUBST for uncertain robotic systems subject to intermittent constraints. The main contributions of the proposed approach are as follows:

(i) Unlike the prescribed-time control scheme based on time-varying unbounded gains in [24], [25], and [26], this paper achieves practical fixed-time stability with PUBST for uncertain robotic systems by employing a bounded time-varying gain. Moreover, compared with the bounded-gain approach in [30], the proposed method relaxes the assumptions on system uncertainties, enabling the control scheme to handle more general system uncertainties and unknown bounded disturbances.

(ii) To alleviate the stringent requirement on prior knowledge of future constraint activation instants in existing intermittent-constrained control approaches [11], [12], [13], [14], a novel stretch function is developed. By introducing transition intervals, the proposed scheme no longer requires the next constraint activation instant to be available upon entering an unconstrained interval and instead only relies on such information shortly before the subsequent constrained phase, thereby reducing conservativeness.

(iii) To address the issue of computational complexity explosion in the derivatives of virtual controllers caused by the stretch function, this paper introduces command filtering technology and designs a filtering error compensation system that achieves practical fixed-time stability with a prescribed upper bound on the settling time (FxTS-PUBST).

Notation: $\|\cdot\|$ denotes the standard Euclidean norm. For $A(t)$, $A(t) \in C^n$ denotes that it has continuous derivatives of order n. For $B = [B_1, B_2, \ldots, B_n]^T \in \mathbb{R}^n$, $|B|^k = [|B_1|^k, |B_2|^k, \ldots, |B_n|^k]^T$ and $\text{sgn}(B) = \text{diag}(\text{sgn}(B_1), \text{sgn}(B_2), \ldots, \text{sgn}(B_n))$.

II. Problem Establishment and Preliminaries

A. System Model

The dynamics of an n-link rigid robotic system can be described as [38]:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau + d, \tag{1}$$

where q, $\dot{q}$, and $\ddot{q} \in \mathbb{R}^n$ are the joint position, velocity, and acceleration, respectively. $\tau = [\tau_1, \tau_2, \ldots, \tau_n] \in \mathbb{R}^n$ denotes the control torque. $M(q) \in \mathbb{R}^{n\times n}$ is the symmetric inertia matrix. $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ denotes the centripetal–coriolis matrix. $G(q) \in \mathbb{R}^n$ represents the gravitational torque. $d \in \mathbb{R}^n$ is the external disturbance. Note that $M(q)$, $C(q,\dot{q})$, $G(q)$, and d are all unknown but satisfy the following properties and assumption.

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Property 1: [39] The matrices $M(q)$, $C(q,\dot{q})$ and the gravitational term $G(q)$ are bounded by

$$\underline{M} \le \|M(q)\| \le \bar{M}, \quad \|C(q,\dot{q})\| \le \bar{C}\|\dot{q}\|, \quad \|G(q)\| \le \bar{G},$$

where $\bar{M}$ is a known positive constant. $\underline{M}$, $\bar{C}$, and $\bar{G}$ are unknown and positive constants.

Property 2: [20] The matrix $\dot{M}(q) - 2C(q,\dot{q})$ is skew symmetric.

Assumption 1: The external disturbance satisfies $\|d\| \le \bar{d}$, where $\bar{d}$ is an unknown positive constant.

Assumption 2: The desired position q_d, velocity $\dot{q}_d$, and acceleration $\ddot{q}_d$ are continuous and uniformly bounded.

Assumption 3: The onboard sensing system can determine the upcoming constraint activation instant $t_{r,j}$ with a minimum preview time of $\Delta T_{\min}$ before entering any constrained environment.

Control objective: Design an event-triggered controller u for system (1), such that all signals of the closed-loop system are bounded, q tracks q_d with the tracking error $e = q - q_d = [e_1, e_2, \ldots, e_n]^T \in \mathbb{R}^n$ converging to a small neighborhood of the origin within the PT T, and e satisfies the following condition:

$$\rho_{d,i} < e_i < \rho_{u,i}, \forall t \in (t_{r,j}, t_{e,j}), \tag{2}$$

where $\rho_{u,i} > 0$ and $\rho_{d,i} < 0$ denote the time-varying upper and lower bounding functions for e_i, respectively, $i = 1, 2, \ldots, n$. $(t_{r,j}, t_{e,j})$ represent the jth constraint interval of e_i, where $0 \le t_{r,j} < t_{e,j}$ with $j = 1, 2, \ldots, N$. Before each constraint interval, the tracking-error constraints are gradually introduced over $(t_{r,j} - \Delta T_1, t_{r,j})$. After the constraint interval, the constraints are gradually relaxed over $(t_{e,j}, t_{e,j} + \Delta T_2)$. Here, ΔT_1 and ΔT_2 are positive constants, with $0 < \Delta T_1 < \Delta T_{\min}$.

Remark 1: Property 1, 2 and Assumption 1 are quite common in the control schemes for uncertain robotic systems [20], [38], [39], [40], [41].

Remark 2: Assumption 3 requires the exact upcoming constraint activation instant $t_{r,j}$ to be available at the beginning of the preactivation transition interval, rather than at the beginning of the unconstrained phase. Therefore, the proposed approach relaxes the requirement for prior knowledge of future constraint activation instants compared with existing intermittent constraint control methods. If the available preview time is shorter than $\Delta T_{\min}$, the theoretical guarantees no longer directly apply. In particular, if the preview time is shorter than ΔT_1, timely completion of the designed transition cannot be guaranteed. Instantaneous switching without sufficient advance notice is outside the scope of this work.

B. Basic Lemmas

Lemma 1: [42] For real variables Υ_1, Υ_2, and the positive constants l_o, $o = 1, 2, 3$, it holds

$$|\Upsilon_1|^{l_1}|\Upsilon_2|^{l_2} \le \frac{l_1}{l_1 + l_2} l_3 |\Upsilon_1|^{l_1+l_2} + \frac{l_2}{l_1 + l_2} l_3^{-\frac{l_1}{l_2}} |\Upsilon_2|^{l_1+l_2}.$$

Lemma 2: [43] For $\aleph \in R$, $\vartheta > 0$, it follows that:

$$(\aleph_1 + \aleph_2 + \ldots + \aleph_n)^{\vartheta} \le \max\{n^{\vartheta-1}, 1\}(\aleph_1^{\vartheta} + \aleph_2^{\vartheta} + \ldots + \aleph_n^{\vartheta}).$$

Lemma 3 [30]: Assuming there exists a Lyapunov function $V(x)$, the following inequality is valid:

$$\dot{V}(x) \le -\varsigma_1 \mu V^p(x) - \varsigma_2 \mu V^q(x), \tag{3}$$

where $\varsigma_1, \varsigma_2 > 0$, $0 < p < 1$ and $q > 1$. The time-varying gain function μ is defined as follows:

$$\mu = \begin{cases} \dfrac{T}{T - t}, & 0 \le t < T_s, \\ -\varrho_1 e^{-(t-T_s)} + \varrho_2, & t \ge T_s \end{cases} \tag{4}$$

where $\varrho_1 = \frac{T}{(T-T_s)^2}$ and $\varrho_2 = \frac{T(T-T_s+1)}{(T-T_s)^2}$. Here, T is the PT and T_s is the switching time. T can be arbitrarily specified, while T_s needs to satisfy

$$T\left(1 - \exp\left\{-\left(\frac{(\varsigma_1/\varsigma_2)^{\frac{1-p}{q-p}}}{(1-p)\varsigma_1 T} + \frac{(\varsigma_1/\varsigma_2)^{\frac{1-q}{q-p}}}{(q-1)\varsigma_2 T}\right)\right\}\right) < T_s < T. \tag{5}$$

Then, we can conclude that the equilibrium point of closed-loop system $\dot{x} = f(x, \tau)$ is FxTS-PUBST.

C. Key Functions

The following key functions play important roles in control design.

1) Stretch Function

The stretch function was first introduced in [11] to establish a continuously differentiable barrier function for all stages, thereby enabling the development of a unified controller for the system with intermittent constraints. However, the stretch function proposed in [11], [12], [13], and [14] requires obtaining information on the next constrained instant at the moment of switching to the unconstrained interval, which is not practical for unpredictable complex dynamic environments. For example, in [13], the stretch function is defined as

$$\check{S}(t) = \begin{cases} e^{\check{S}_0(t)}, & t \in [0, t_{r,1}) \\ 1, & t \in [t_{r,j}, t_{e,j}), \ j = 1, 2, \ldots, N \\ e^{\check{S}_g(t)}, & t \in [t_{e,g}, t_{r,g+1}), \ g = 1, 2, \ldots, N-1 \\ e^{\check{S}_N(t)}, & t \in [t_{e,N}, +\infty) \end{cases} \tag{6}$$

with

$$\check{S}_0(t) = -\frac{(t - t_{r,1})^{2n}}{\sigma_0}, \check{S}_N(t) = -\frac{(t_{e,N} - t)^{2n}}{\sigma_N},$$
$$\check{S}_g(t) = \left[\frac{-2n\left(2t - t_{e,g} - t_{r,g+1}\right)^{2n+1}}{\left(t_{r,g+1} - t_{e,g}\right)^{2n+1}}\mathrm{sgn}(\cdot) + \frac{(2n+1)\left(2t - t_{e,g} - t_{r,g+1}\right)^{2n}}{\left(t_{r,g+1} - t_{e,g}\right)^{2n}} - 1\right] \Big/ \sigma_g,$$

where $\mathrm{sgn}(\cdot) = \mathrm{sgn}(2t - t_{e,g} - t_{r,g+1}), 0 < \sigma_i \ll 1 (i = 0, 1, \ldots, N)$. Here, $t_{r,j}$ and $t_{e,j}$ denote the instants when the system enters and leaves the confined space for the jth time, respectively. From (6), it can be concluded that $t_{r,1}$ is known at time $t = 0$.

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Similarly, at time $t = t_{e,g}$, $t_{r,g+1}$ should be available. To address this issue, we propose an improved stretch function as follows:

$$S(t) = \begin{cases} 0, & t \in [0, \underline{t}_{r,1}], \\ S_{1,j}(t), & t \in (\underline{t}_{r,j}, t_{r,j}], \\ 1, & t \in (t_{r,j}, t_{e,j}), \\ S_{2,j}(t), & t \in [t_{e,j}, t_{e,j} + \Delta T_2), \\ 0, & t \in [t_{e,g} + \Delta T_2, \underline{t}_{r,g+1}], \\ 0, & t \in [t_{e,N} + \Delta T_2, \infty) \end{cases} \tag{7}$$

with

$$S_{1,j}(t) = e^{-\left(\frac{1}{t-\underline{t}_{r,j}} - \frac{1}{\Delta T_1}\right)^3}, \quad S_{2,j}(t) = e^{-\left(\frac{t-t_{e,j}}{\Delta T_2 - t + t_{e,j}}\right)^3},$$

where $\underline{t}_{r,j} = t_{r,j} - \Delta T_1$. $g = 1, 2, \ldots, N-1$, and $j = 1, 2, \ldots, N$. Define

$$\begin{aligned} \mathcal{I}_{\text{pre},j} &= (\underline{t}_{r,j}, t_{r,j}], \\ \mathcal{I}_{\text{c},j} &= (t_{r,j}, t_{e,j}), \\ \mathcal{I}_{\text{post},j} &= [t_{e,j}, t_{e,j} + \Delta T_2), \\ \mathcal{I}_{\text{f},j} &= \begin{cases} [t_{e,j} + \Delta T_2, \underline{t}_{r,j+1}], & j = 1, \ldots, N-1, \\ [t_{e,N} + \Delta T_2, \infty), & j = N. \end{cases} \end{aligned} \tag{8}$$

Here, $\mathcal{I}_{\text{pre},j}$ and $\mathcal{I}_{\text{post},j}$ denote the "transition intervals" before the system enters and after it leaves the confined space for the jth time, respectively. $\mathcal{I}_{\text{c},j}$ and $\mathcal{I}_{\text{f},j}$ denote the constrained and unconstrained time intervals, respectively.

$S(t)$ has the following properties:

P_1.$S(t) \in [0, 1]$ for $t \geq 0$.

P_2.$S(t) \in C^2$.

Proof: By taking the time-derivatives of $S_{1,j}(t)$, it can be concluded that

$$\dot{S}_{1,j}(t) = 3\left(\frac{1}{t-\underline{t}_{r,j}} - \frac{1}{\Delta T_1}\right)^2 (t-\underline{t}_{r,j})^{-2} e^{-\left(\frac{1}{t-\underline{t}_{r,j}} - \frac{1}{\Delta T_1}\right)^3}. \tag{9}$$

Then, it can be readily derived that $S_{1,j}(t)$ is monotonically increasing, $\lim_{t\to\underline{t}_{r,j}^+} S_{1,j}(t) = 0$ and $S_{1,j}(t_{r,j}) = 1$. Similarly, we have

$$\dot{S}_{2,j}(t) = -3\Delta T_2 \frac{(t-t_{e,j})^2}{(\Delta T_2 - t + t_{e,j})^4} e^{-\left(\frac{t-t_{e,j}}{\Delta T_2 - t + t_{e,j}}\right)^3}, \tag{10}$$

$S_{2,j}(t)$ is monotonically decreasing, $S_{2,j}(t_{e,j}) = 1$ and $\lim_{t\to t_{e,j}+\Delta T_2^-} S_{2,j}(t) = 0$. Moreover, combining (7), we know that $S(t) \in [0, 1]$ for $t \geq 0$.

From (7), it is readily obtained that $S(t)$ is continuous, and second-order differentiable except $t = \underline{t}_{r,j}$, $t_{r,j}$, $t_{e,j}$ and $t_{e,j} + \Delta T_2$.

According to (9) and (10), one has

$$\begin{aligned} \ddot{S}_{1,j}(t) = &-3\left(\frac{1}{t-\underline{t}_{r,j}} - \frac{1}{\Delta T_1}\right)(t-\underline{t}_{r,j})^{-3}[2(t-\underline{t}_{r,j})^{-1} \\ &+ 2\left(\frac{1}{t-\underline{t}_{r,j}} - \frac{1}{\Delta T_1}\right) - 3\left(\frac{1}{t-\underline{t}_{r,j}} - \frac{1}{\Delta T_1}\right)^3 \\ &\times (t-\underline{t}_{r,j})^{-1}]e^{-\left(\frac{1}{t-\underline{t}_{r,j}} - \frac{1}{\Delta T_1}\right)^3}, \end{aligned} \tag{11}$$

and

$$\begin{aligned} \ddot{S}_{2,j}(t) = &-3\Delta T_2 \frac{t-t_{e,j}}{(\Delta T_2 - t + t_{e,j})^4}\Big[2 + \frac{4(t-t_{e,j})}{\Delta T_2 - t + t_{e,j}} \\ &- 3\frac{\Delta T_2 (t-t_{e,j})^3}{(\Delta T_2 - t + t_{e,j})^4}\Big] e^{-\left(\frac{t-t_{e,j}}{\Delta T_2 - t + t_{e,j}}\right)^3}. \end{aligned} \tag{12}$$

An obvious conclusion is that $\dot{S}_{1,j}(t_{r,j}) = \lim_{t\to t_{r,j}^+} \dot{S}(t) = 0$, $\ddot{S}_{1,j}(t_{r,j}) = \lim_{t\to t_{r,j}^+} \ddot{S}(t) = 0$, $\lim_{t\to t_{e,j}^-} \dot{S}(t) = \dot{S}_{2,j}(t_{e,j}) = 0$ and $\lim_{t\to t_{e,j}^-} \ddot{S}(t) = \ddot{S}_{2,j}(t_{e,j}) = 0$.

Considering the dominant decay of the exponential term, we have $\lim_{t\to\underline{t}_{r,j}^+} \dot{S}_{1,j}(t) = \dot{S}(\underline{t}_{r,j}) = 0$, $\lim_{t\to\underline{t}_{r,j}^+} \ddot{S}_{1,j}(t) = \ddot{S}(\underline{t}_{r,j}) = 0$, $\lim_{t\to t_{e,j}+\Delta T_2^-} \dot{S}_{2,j}(t) = \dot{S}(t_{e,j} + \Delta T_2) = 0$, and $\lim_{t\to t_{e,j}+\Delta T_2^-} \ddot{S}_{2,j}(t) = \ddot{S}(t_{e,j} + \Delta T_2) = 0$.

Therefore, $S(t) \in C^2$. □

2) Barrier Function

Considering the following transformation:

$$\delta_i(t) = S(t)e_i(t), \quad i = 1, 2, \ldots, n. \tag{13}$$

Meanwhile, design a stretch function-based barrier function as follows:

$$\varpi_i = \frac{\rho_{u,i}\rho_{d,i}e_i}{(\rho_{u,i} - \delta_i)(\rho_{d,i} - \delta_i)}, \tag{14}$$

The derivative of the barrier function (14) is expressed as follows:

$$\dot{\varpi}_i = \psi_i \dot{e}_i + \phi_i, \tag{15}$$

where

$$\psi_i = \frac{(C_i - \delta_i^2)C_i}{A_i^2 B_i^2} > 0, \tag{16}$$

$$\phi_i = \frac{A_i B_i \dot{C}_i e_i - D_{A,i} B_i C_i e_i - A_i D_{B,i} C_i e_i}{A_i^2 B_i^2}, \tag{17}$$

and $A_i = \rho_{u,i} - \delta_i$, $B_i = \rho_{d,i} - \delta_i$, $C_i = \rho_{u,i}\rho_{d,i}$, $D_{A,i} = \dot{\rho}_{u,i} - \dot{S}e_i$, $D_{B,i} = \dot{\rho}_{d,i} - \dot{S}e_i$. ψ_i is well-defined and $\psi_i > 0$ for $\rho_{d,i} < \delta_i < \rho_{u,i}$.

Let $\varpi = [\varpi_1, \varpi_2, \ldots, \varpi_n]^T$. Then, we can obtain

$$\dot{\varpi} = \psi \dot{e} + \phi, \tag{18}$$

where $\psi = \text{diag}(\psi_1, \psi_2, \ldots, \psi_n)$, $\phi = [\phi_1, \phi_2, \ldots, \phi_n]^T$.

3) Switch Function

$$\Theta(\varkappa_i) = \begin{cases} 1, & |\varkappa_i| \geq \varepsilon \\ \sin^2\left(\frac{\varkappa_i \pi}{2\varepsilon}\right), & |\varkappa_i| < \varepsilon \end{cases} \tag{19}$$

and

$$\Theta_d(\varkappa) = \text{diag}(\Theta(\varkappa_1), \Theta(\varkappa_2), \ldots, \Theta(\varkappa_n)), \tag{20}$$

where $\varkappa = [\varkappa_1, \varkappa_2, \ldots, \varkappa_n] \in \mathbb{R}^n$, ε is a positive design parameter and $i = 1, 2, \ldots, n$.

Remark 3: The instant $\underline{t}_{r,j}$, $j = 1, 2, \ldots, N$, represents the time at which the system detects that the trajectory is about to enter a constrained environment. After $\underline{t}_{r,j}$, the controller is gradually switched to a constraint-handling mode over the interval ΔT_1 to ensure the desired tracking accuracy. The parameters ΔT_1 and ΔT_2 are prespecified transition intervals introduced to guarantee smooth switching.

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III. Main Results

A. Practical Fixed-Time Stability Theorem

In this section, a novel practical fixed-time stability theorem with prescribed upper bound of the settling-time is proposed.

Before introducing this theory, we first analyze some properties of the time-varying gain function μ. Based on (4), it can be concluded that

$$\dot{\mu} = \begin{cases} \dfrac{T}{(T-t)^2}, & 0 \le t < T_s, \\ \varrho_1 e^{-(t-T_s)}, & t \ge T_s. \end{cases} \tag{21}$$

Then, a direct conclusion yields that μ is monotonically increasing and $1 \le \mu < \varrho_2$. To facilitate the scaling transformation of μ, let $\underline{\mu} = 1$ and $\bar{\mu} = \varrho_2$.

Theorem 1: If there exists a Lyapunov function $V(x)$ such that any solution x of the closed-loop system $\dot{x} = f(x, \tau)$ satisfies the inequality

$$\dot{V}(x) \le -\varsigma_1 \mu V^p(x) - \varsigma_2 \mu V^q(x) + \Xi \tag{22}$$

for the time-varying function μ defined in equation (4) and some constants $\varsigma_1, \varsigma_2 > 0$, $0 < p < 1$, $q > 1$ and $0 < \Xi < \infty$, then the solution of the system is practical FxTS-PUBST.

Proof: By introducing a constant $\lambda \in (0, 1)$, (22) can be rewritten as

$$\dot{V}(x) \le -\lambda\varsigma_1 \mu V^p(x) - (1-\lambda)\varsigma_1 \mu V^p(x) - \varsigma_2 \mu V^q(x) + \Xi \tag{23}$$

or

$$\dot{V}(x) \le -\varsigma_1 \mu V^p(x) - \lambda\varsigma_2 \mu V^q(x) - (1-\lambda)\varsigma_2 \mu V^q(x) + \Xi. \tag{24}$$

For (23), if $V^p(x) > \Xi/(1-\lambda)\varsigma_1$, it follows that:

$$\dot{V}(x) \le -\lambda\varsigma_1 \mu V^p(x) - \varsigma_2 \mu V^q(x). \tag{25}$$

Further, from Lemma 3, the decrease of $V(x)$ in PT T drives the solution of the system into $V^p(x) \le \Xi/(1-\lambda)\varsigma_1\underline{\mu}$ and T_s needs to satisfy

$$\underline{T}_{s_1} < T_s < T, \tag{26}$$

where

$$\underline{T}_{s_1} = T\left(1 - \exp\left\{-\left(\frac{(\lambda\varsigma_1/\varsigma_2)^{\frac{1-p}{q-p}}}{(1-p)\lambda\varsigma_1 T} + \frac{(\lambda\varsigma_1/\varsigma_2)^{\frac{1-q}{q-p}}}{(q-1)\varsigma_2 T}\right)\right\}\right). \tag{27}$$

Similarly, by performing the above analysis on (24), we can obtain the decrease of $V(x)$ in the PT T drives the solution of the system into $V^q(x) \le \Xi/(1-\lambda)\varsigma_2$ and T_s needs to satisfy

$$\underline{T}_{s_2} < T_s < T, \tag{28}$$

where

$$\underline{T}_{s_2} = T\left(1 - \exp\left\{-\left(\frac{(\varsigma_1/\lambda\varsigma_2)^{\frac{1-p}{q-p}}}{(1-p)\varsigma_1 T} + \frac{(\varsigma_1/\lambda\varsigma_2)^{\frac{1-q}{q-p}}}{(q-1)\lambda\varsigma_2 T}\right)\right\}\right). \tag{29}$$

Therefore, the solution of the system has an adjustable upper boundary for $t \ge T$ as

$$x \in \left\{V(x) \le \min\left\{[\Xi/(1-\lambda)\varsigma_1]^{\frac{1}{p}}, [\Xi/(1-\lambda)\varsigma_2]^{\frac{1}{q}}\right\}\right\} \tag{30}$$

and T_s should satisfy

$$\max\left\{\underline{T}_{s_1}, \underline{T}_{s_2}\right\} < T_s < T. \tag{31}$$

This completes the proof. □

Remark 4 [28]: points out that the PT control algorithms proposed in [23], [24], and [25] suffer from the issue of infinite controller gain. Although these papers address this issue by deactivating the controller prior to reaching the prescribed settling-time, such approaches render the system sensitive to arbitrarily small noise before the prescribed settling-time. However, the new fixed-time Lyapunov theorem proposed in [30], based on the time-varying gain function (4) that is continuously differentiable and uniformly bounded, namely Lemma 3, completely avoids the aforementioned issues. Building on Lemma 3, this paper develops a new practical fixed-time Lyapunov theorem that shares the same advantage.

Remark 5: The bounded time-varying gain function (4) is adopted from Lemma 3 of [30]. Theorem 1 extends the inequality (3) by allowing an additive residual $\Xi > 0$. This extension supports the stability analysis of robotic systems with model uncertainties and unknown bounded disturbances. To account for the additive residual, Theorem 1 provides explicit residual-set bounds and derives a corresponding sufficient condition on T_s.

B.: Event-Triggered Adaptive Fixed-Time Controller Design

Define the following change of coordinates:

$$\begin{cases} z_1 = \varpi, \\ z_2 = \dot{q} - \bar{\alpha}_2, \end{cases} \tag{32}$$

where $\bar{\alpha}_2 \in \mathbb{R}^n$ denotes the output of the first-order filter.

The form of the first-order filter is given as follows:

$$\xi_2 \dot{\bar{\alpha}}_2 + \bar{\alpha}_2 = \alpha_1, \tag{33}$$

where $\xi_2 > 0$ is a design parameter, and α_1 is the filter input. The initial state is given by $\bar{\alpha}_2(0) = \alpha_1(0)$. To compensate for the errors caused by the filter, the error compensation system is designed as follows:

$$\begin{cases} \dot{\lambda}_1 = \psi(\bar{\alpha}_2 - \alpha_1) + \psi\lambda_2 - \dfrac{b_1}{4}\mu\lambda_1\lambda_1^T\lambda_1 - \dfrac{b_2}{2^p}\mu\Theta_d(\lambda_1) \\ \qquad\qquad \times \mathrm{sgn}(\lambda_1)|\lambda_1|^{2p-1} - \dfrac{1}{2}\psi^T\psi\lambda_1, \\ \dot{\lambda}_2 = -\dfrac{b_3}{4}\mu\lambda_2\lambda_2^T\lambda_2 - \dfrac{b_4}{2^p}\mu\Theta_d(\lambda_2)\mathrm{sgn}(\lambda_2)|\lambda_2|^{2p-1} - \psi\lambda_1, \end{cases} \tag{34}$$

where $\lambda_k = [\lambda_{k,1}, \lambda_{k,2}, \ldots, \lambda_{k,n}] \in \mathbb{R}^n$, $k = 1, 2$, are the states of system (34). b_o, $o = 1, 2, 3, 4$, p are positive design parameters with $0.5 < p < 1$.

The compensated tracking error can be written as follows:

$$\begin{cases} v_1 = z_1 - \lambda_1, \\ v_2 = z_2 - \lambda_2, \end{cases} \tag{35}$$

where $v_k = [v_{k,1}, v_{k,2}, \ldots, v_{k,n}] \in \mathbb{R}^n$, $k = 1, 2$.

Prior to designing the controller, the following key variables are defined $\theta_1 = \bar{M}^2$, $\theta_2 = \bar{C}^2$, $\theta_3 = \bar{G}^2$, $\theta_4 = \bar{d}^2$, $\theta = \max\{\theta_1, \theta_2, \theta_3, \theta_4\}$ and $\tilde{\theta} = \theta - \hat{\theta}$. $\hat{\theta}$ denotes the estimate of θ. The adaptive law $\dot{\hat{\theta}}$ and the virtual controller α_1, α_2 are constructed as

$$\dot{\hat{\theta}} = \frac{r}{2}v_2^T v_2 \eta - \mu h_1 \hat{\theta} - \mu\frac{h_2}{r}\hat{\theta}^3, \tag{36}$$

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$$\alpha_1 = -\psi^{-1}\left(\frac{a_1}{4}\mu v_1 v_1^T v_1 + \frac{a_2}{2^p}\mu\Theta_d(v_1)\mathrm{sgn}(v_1)|v_1|^{2p-1} + \phi + \frac{b_1}{4}\mu\lambda_1\lambda_1^T\lambda_1 + \frac{b_2}{2^p}\mu\Theta_d(\lambda_1)\mathrm{sgn}(\lambda_1)|\lambda_1|^{2p-1} + \frac{1}{2}\psi^T\psi\lambda_1\right) + \dot{q}_d, \tag{37}$$

$$\alpha_2 = -\frac{a_3}{4}\mu v_2 v_2^T v_2 - \frac{a_4}{2^p}\mu\Theta_d(v_2)\mathrm{sgn}(v_2)|v_2|^{2p-1} - \psi v_1 - \frac{1}{2}v_2\hat{\theta}\eta, \tag{38}$$

where r, h_k and a_o, $k = 1, 2$, $o = 1, 2, 3, 4$, are positive design parameters. $\alpha_k = [\alpha_{k,1}, \alpha_{k,2}, \ldots, \alpha_{k,n}] \in \mathbb{R}^n$. η is a time-varying function with its form provided later.

We propose the following control design procedure.

***Step 1*:** According to (1), (32) and (35), the derivative of v_1 is formulated as

$$\begin{aligned}\dot{v}_1 &= \dot{z}_1 - \dot{\lambda}_1 \\ &= \psi\dot{q} - \psi\dot{q}_d + \phi - \dot{\lambda}_1.\end{aligned} \tag{39}$$

Choose the Lyapunov function candidate as

$$V_1 = \frac{1}{2}v_1^T v_1. \tag{40}$$

Utilizing (39), the derivative of V_1 is expressed as follows:

$$\dot{V}_1 = v_1^T\left(\psi v_2 + \psi\alpha_1 - \psi\dot{q}_d + \phi + \psi\bar{\alpha}_2 + \psi\lambda_2 - \psi\alpha_1 - \dot{\lambda}_1\right). \tag{41}$$

Substituting (34) and (37) into (41) obtains

$$\dot{V}_1 = -a_1\mu\left(\frac{1}{2}v_1^T v_1\right)^2 - a_2\mu\sum_{i=1}^{n}\Theta(v_{1,i})\left(\frac{1}{2}v_{1,i}^2\right)^p + v_1^T\psi v_2. \tag{42}$$

***Step 2*:** Define

$$V_2 = \frac{1}{2}v_2^T M v_2. \tag{43}$$

Taking the derivative of (43), we have

$$\dot{V}_2 = v_2^T M\dot{v}_2 + \frac{1}{2}v_2^T\dot{M}v_2. \tag{44}$$

Using Property 2 yields

$$\begin{aligned}\dot{V}_2 &= v_2^T M\dot{v}_2 + v_2^T C v_2 + \frac{v_2^T(\dot{M} - 2C)v_2}{2} \\ &= v_2^T M\dot{v}_2 + v_2^T C v_2.\end{aligned} \tag{45}$$

Consider the following equation

$$\begin{aligned}M\dot{v}_2 + Cv_2 &= M(\ddot{q} - \dot{\bar{\alpha}}_2 - \dot{\lambda}_2) + C(\dot{q} - \bar{\alpha}_2 - \lambda_2) \\ &= \tau + d - G - M(\dot{\bar{\alpha}}_2 + \dot{\lambda}_2) - C(\bar{\alpha}_2 + \lambda_2).\end{aligned} \tag{46}$$

Substituting (46) into (45) obtains

$$\dot{V}_2 = v_2^T\left[\tau + d - G - M(\dot{\bar{\alpha}}_2 + \dot{\lambda}_2) - C(\bar{\alpha}_2 + \lambda_2)\right]. \tag{47}$$

Utilizing Young's inequality, Property 1 and Assumption 1 , we obtain

$$\dot{V}_2 \le v_2^T\left[(\tau - \alpha_2) + \alpha_2 + \frac{1}{2}v_2\theta\eta\right] + 2, \tag{48}$$

where $\eta = \|\dot{\bar{\alpha}}_2 + \dot{\lambda}_2\|^2 + \|\dot{q}\|^2\|\bar{\alpha}_2 + \lambda_2\|^2 + 2 > 0$.

By (38), one has

$$\dot{V}_2 \le -a_3\mu\left(\frac{1}{2}v_2^T v_2\right)^2 - a_4\mu\sum_{i=1}^{n}\Theta(v_{2,i})\left(\frac{1}{2}v_{2,i}^2\right)^p - v_2^T\psi v_1 + \frac{1}{2}v_2^T v_2\tilde{\theta}\eta + v_2^T(\tau - \alpha_2) + 2. \tag{49}$$

***Step 3*:** Choose the Lyapunov function candidate as

$$V_3 = V_1 + V_2 + V_\theta, \tag{50}$$

where $V_\theta = \frac{1}{2r}\tilde{\theta}^2$.

Taking the derivative of (50), we have

$$\begin{aligned}\dot{V}_3 &= \dot{V}_1 + \dot{V}_2 + \dot{V}_\theta \\ &\le -\sum_{k=1}^{2}\left[a_{2k-1}\mu\left(\frac{1}{2}v_k^T v_k\right)^2 + a_{2k}\mu\sum_{i=1}^{n}\Theta(v_{k,i})\left(\frac{1}{2}v_{k,i}^2\right)^p\right] + 2 - \frac{1}{r}\tilde{\theta}\left(\dot{\hat{\theta}} - \frac{r}{2}v_2^T v_2\eta\right) + v_2^T(\tau - \alpha_2).\end{aligned} \tag{51}$$

By (36), we have

$$-\frac{1}{r}\tilde{\theta}\left(\dot{\hat{\theta}} - \frac{r}{2}v_2^T v_2\eta\right) = \mu\frac{h_1}{r}\tilde{\theta}\hat{\theta} + \mu\frac{h_2}{r^2}\tilde{\theta}\hat{\theta}^3. \tag{52}$$

Based on Lemma 1, we have

$$\mu\frac{h_1}{2r}\tilde{\theta}^2 \ge \mu h_1\left(\frac{1}{2r}\tilde{\theta}^2\right)^p - \mu h_1(1-p)p^{\frac{p}{1-p}}, \tag{53}$$

$$\mu\frac{h_2}{r^2}\tilde{\theta}\theta^3 \le \mu\frac{3h_2}{r^2}\tilde{\theta}^2\theta^2 + \mu\frac{h_2}{12r^2}\theta^4, \tag{54}$$

$$\mu\frac{3h_2}{r^2}\tilde{\theta}^3\theta \le \mu\frac{9\chi^{\frac{4}{3}}h_2}{4r^2}\tilde{\theta}^4 + \mu\frac{3h_2}{4r^2\chi^4}\theta^4, \tag{55}$$

where χ is a positive constant.

Substituting (52)-(55) into (51), we obtain

$$\dot{V}_3 \le -\sum_{k=1}^{2}\left[a_{2k-1}\mu\left(\frac{1}{2}v_k^T v_k\right)^2 + a_{2k}\mu\sum_{i=1}^{n}\Theta(v_{k,i})\left(\frac{1}{2}v_{k,i}^2\right)^p\right] - \mu h_1\left(\frac{1}{2r}\tilde{\theta}^2\right)^p - \mu\left(4h_2 - 9\chi^{\frac{4}{3}}h_2\right)\left(\frac{1}{2r}\tilde{\theta}^2\right)^2 + D_1 + v_2^T(\tau - \alpha_2). \tag{56}$$

where

$$D_1 = 2 + \bar{\mu}\frac{h_1}{2r}\theta^2 + \bar{\mu}\left(\frac{3h_2}{4r^2\chi^4} + \frac{h_2}{12r^2}\right)\theta^4 + \bar{\mu}h_1(1-p)p^{\frac{p}{1-p}}. \tag{57}$$

To mitigate communication burdens, the event-triggered mechanism is designed such that

$$\begin{cases}\zeta_i(t) = -(1+\sigma_i)\left[\alpha_{2,i}\tanh\left(\frac{v_{2,i}\alpha_{2,i}}{\iota_i}\right) + \omega_i\tanh\left(\frac{v_{2,i}\omega_i}{\iota_i}\right)\right], \\ \tau_i(t) = \zeta_i(t_{k,i}), \quad \forall t \in [t_{k,i}, t_{k+1,i}), \\ t_{k+1,i} = \inf\{t > t_{k,i}\,|\,|E_{k+1,i}(t)| \ge \sigma_i|\tau_i| + \omega_i\},\end{cases} \tag{58}$$

where $\sigma_i < 1$, ι_i and ω_i are positive design parameters and $E_{k+1,i}(t) = \zeta_i(t) - \tau_i(t)$. $i = 1, 2, \ldots, n$.

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From (58), it follows that during $t \in [t_{k,i}, t_{k+1,i})$, $e_{k+1,i}(t)$ satisfies

$$|E_{k+1,i}(t)| \le \sigma_i|\tau_i| + \omega_i. \tag{59}$$

Consider two continuous time-varying functions $\Lambda_1(t)$ and $\Lambda_2(t)$ that satisfy $|\Lambda_1(t)| < 1$, $|\Lambda_2(t)| < 1$, $|\Lambda_1(t_{k,i})| = |\Lambda_2(t_{k,i})| = 0$ and $|\Lambda_1(t_{k+1,i})| = |\Lambda_2(t_{k+1,i})| = 1$ for $t \in [t_{k,i}, t_{k+1,i})$.

Based on (59) and the properties of $\Lambda_1(t)$ and $\Lambda_2(t)$, one has

$$\zeta_i(t) - \tau_i(t) = \sigma_i\Lambda_1(t)\tau_i(t) + \Lambda_2(t)\omega_i. \tag{60}$$

Thus

$$\tau_i(t) = \frac{\zeta_i(t) - \Lambda_2(t)\omega_i}{1 + \sigma_i\Lambda_1(t)}. \tag{61}$$

Combining (58) and (61) yields

$$\begin{aligned}\tau_i(t) = -\frac{1+\sigma_i}{1+\sigma_i\Lambda_1(t)}\Bigg[\alpha_{2,i}\tanh\left(\frac{v_{2,i}\alpha_{2,i}}{\iota_i}\right) \\ +\omega_i\tanh\left(\frac{v_{2,i}\omega_i}{\iota_i}\right)\Bigg] - \frac{\Lambda_2(t)\omega_i}{1+\sigma_i\Lambda_1(t)}.\end{aligned} \tag{62}$$

By Lemma 3 of [44], one has

$$\begin{aligned}v_{2,i}\tau_i(t) &\le -\frac{1+\sigma_i}{1+\sigma_i\Lambda_1(t)}|v_{2,i}^T\alpha_{2,i}| - \frac{1+\sigma_i}{1+\sigma_i\Lambda_1(t)} \\ &\quad\times|v_{2,i}\omega_i| + \left|\frac{\Lambda_2(t)v_{2,i}\omega_i}{1+\sigma_i\Lambda_1(t)}\right| + 0.557\frac{1+\sigma_i}{1+\sigma_i\Lambda_1(t)}\iota_i \\ &\le v_{2,i}\alpha_{2,i} + 0.557\frac{1+\sigma_i}{1-\sigma_i}\iota_i.\end{aligned} \tag{63}$$

Furthermore, we can obtain

$$v_2\tau \le v_2\alpha_2 + 0.557\sum_{i=1}^{n}\frac{1+\sigma_i}{1-\sigma_i}\iota_i. \tag{64}$$

Substituting (64) into (56) yields

$$\begin{aligned}\dot V_3 \le &-\sum_{k=1}^{2}\left[a_{2k-1}\mu\left(\frac{1}{2}v_k^Tv_k\right)^2 + a_{2k}\mu\sum_{i=1}^{n}\Theta(v_{k,i})\left(\frac{1}{2}v_{k,i}^2\right)^p\right] \\ &-\mu h_1\left(\frac{1}{2r}\tilde\theta^2\right)^p - \mu\left(4h_2 - 9\chi^{\frac{4}{3}}h_2\right)\left(\frac{1}{2r}\tilde\theta^2\right)^2 \\ &+ D_2,\end{aligned} \tag{65}$$

where $D_2 = D_1 + 0.557\sum_{i=1}^{n}\frac{1+\sigma_i}{1-\sigma_i}\iota_i$.

Remark 6: The quartic adaptive-error coefficient obtained from inequality (56) is

$$c_\theta(\chi) = h_2(4 - 9\chi^{4/3}). \tag{66}$$

Since $h_2 > 0$, positivity is equivalent to $0 < \chi < \chi_{\max}$, where $\chi_{\max} = (4/9)^{3/4}$. A convenient parametrization is $\chi = \kappa\chi_{\max}$ with $0 < \kappa < 1$, which yields

$$\begin{aligned}c_\theta &= 4h_2(1 - \kappa^{4/3}), \\ D_\chi &= \frac{3\bar\mu h_2\theta^4}{4r^2\chi_{\max}^4}\kappa^{-4}.\end{aligned} \tag{67}$$

Here, D_χ is the χ-dependent part of D_1, and hence also of D_3 and D. Smaller κ increases c_θ but enlarges the residual estimate as κ^{-4}; as $\kappa \to 1^-$, $c_\theta \to 0$ and the coefficient A can become small. Therefore, neither endpoint is a suitable choice. One may start with $\kappa = 1/2$, then check A and D. This is an admissible starting value, rather than an optimal choice. Since χ appears only in the Young-inequality estimate and not in the implemented controller, changing it affects the bounds in the stability analysis rather than the simulated trajectory.

Remark 7: According to (1), (13)-(18), (32)-(36), Assumption 2 and P_2, it can be concluded that $\dot\psi$, $\dot\mu$, $\dot\nu_k$, $\dot\phi$, $\dot\lambda_k$, $\dot{\bar q}_d$, $k = 1, 2$, exist. To avoid the singularity problem of $\dot\alpha_1$, $\dot\alpha_2$, the switch function $\Theta(\varkappa)$ is introduced. Let

$$\mho(\varkappa_i) = \Theta(\varkappa_i)\mathrm{sgn}(\varkappa_i)|\varkappa_i|^{2p-1}. \tag{68}$$

If $\Theta(\varkappa_i) = 1$, then $\mho(\varkappa_i) = \mathrm{sgn}(\varkappa_i)|\varkappa_i|^{2p-1}$,

$$\dot\mho(\varkappa_i) = \begin{cases}(2p-1)\varkappa_i^{2p-2}\dot\varkappa_i, & \varkappa_i \ge 0, \\ (2p-1)(-\varkappa_i)^{2p-2}\dot\varkappa_i, & \varkappa_i < 0\end{cases} \tag{69}$$

and $\lim_{\varkappa_i\to 0}\dot\mho(\varkappa_i) = \infty$, which creates the singularity problem of $\dot\alpha_1$, and $\dot\alpha_2$. Based on (19) and (68), it is easy to show that $\mho(\varkappa_i)$ is differentiable except for $\varkappa_i = 0$. Now, we discuss the differentiability of $\mho(\varkappa_i)$ at this point. For $|\varkappa_i| \le \varepsilon$,

$$\dot\mho(\varkappa_i) = \begin{cases}\gamma_1(\varkappa_i)\dot\varkappa_i, & \varkappa_i \ge 0, \\ \gamma_2(\varkappa_i)\dot\varkappa_i, & \varkappa_i < 0\end{cases} \tag{70}$$

where

$$\begin{aligned}\gamma_1(\varkappa_i) = &\frac{\pi}{\varepsilon}\sin\left(\frac{\varkappa_i\pi}{2\varepsilon}\right)\cos\left(\frac{\varkappa_i\pi}{2\varepsilon}\right)\varkappa_i^{2p-1} + (2p-1) \\ &\times\sin^2\left(\frac{\varkappa_i\pi}{2\varepsilon}\right)\varkappa_i^{2p-2}\end{aligned}$$

and

$$\begin{aligned}\gamma_2(\varkappa_i) = &-\frac{\pi}{\varepsilon}\sin\left(\frac{\varkappa_i\pi}{2\varepsilon}\right)\cos\left(\frac{\varkappa_i\pi}{2\varepsilon}\right)(-\varkappa_i)^{2p-1} + (2p-1) \\ &\times\sin^2\left(\frac{\varkappa_i\pi}{2\varepsilon}\right)(-\varkappa_i)^{2p-2}.\end{aligned}$$

Then, by (70) and L'Hôpital's rule, we have that $\lim_{\varkappa_i\to 0^+}\gamma_1(\varkappa_i) = \lim_{\varkappa_i\to 0^-}\gamma_2(\varkappa_i) = 0$, which denotes that $\lim_{\varkappa_i\to 0}\dot\mho(\varkappa_i) = 0$. Thus, $\dot\mho(v)$ is continuous and bounded, and $\mho(\varkappa_i)$ is differentiable at $\varkappa_i = 0$. Therefore, the singularity problem is avoided.

IV. Stability Analysis

Theorem 2: For the n-link rigid robotic system (1) subject to intermittent constraints under Assumptions 1–3 and the condition $0 < \Delta T_1 < \Delta T_{\min}$, using the adaptive law (36), virtual controller (37), (38) and event-triggered mechanism (58) can guarantee that: (i). the closed-loop system is practical FxTS-PUBST; (ii). the tracking error e(t) satisfies intermittent constraints; (iii). all signals in the closed-loop system are bounded. (iv). the Zeno behavior is excluded.

Proof: 1). Consider the Lyapunov function as follows:

$$V = V_3 + V_\lambda, \tag{71}$$

where $V_\lambda = \frac{1}{2}\lambda_1^T\lambda_1 + \frac{1}{2}\lambda_2^T\lambda_2$.

Taking the derivative of (71), one has

$$\dot V = \dot V_3 + \dot V_\lambda. \tag{72}$$

For $\dot V_\lambda$, according to (34), one has

$$\dot V_\lambda = \lambda_1^T\dot\lambda_1 + \lambda_2^T\dot\lambda_2$$

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$$= -b_1\mu\left(\frac{1}{2}\lambda_1^T\lambda_1\right)^2 - b_2\mu\sum_{i=1}^{n}\Theta(\lambda_{1,i})\left(\frac{1}{2}\lambda_{1,i}^2\right)^p - b_3\mu\left(\frac{1}{2}\lambda_2^T\lambda_2\right)^2 - b_4\mu\sum_{i=1}^{n}\Theta(\lambda_{2,i})\left(\frac{1}{2}\lambda_{2,i}^2\right)^p - \frac{1}{2}\lambda_1^T\psi^T\psi\lambda_1 + \lambda_1^T\psi(\bar{\alpha}_2-\alpha_1). \tag{73}$$

According to [45], for positive vector Γ, there exists ξ_2 such that $|\bar{\alpha}_2-\alpha_1|<\Gamma$. Furthermore, using Young's inequality, the last term of (73) can be rewritten as

$$\lambda_1^T\psi(\bar{\alpha}_2-\alpha_1) \le \frac{1}{2}\lambda_1^T\psi^T\psi\lambda_1 + \frac{1}{2}\Gamma^T\Gamma. \tag{74}$$

Then, combining (65), (72), (73) and (74) yields

$$\dot{V} \le -\sum_{k=1}^{2}\left[a_{2k-1}\mu\left(\frac{1}{2}v_k^Tv_k\right)^2 + a_{2k}\mu\sum_{i=1}^{n}\Theta(v_{k,i})\left(\frac{1}{2}v_{k,i}^2\right)^p + b_{2k-1}\mu\left(\frac{1}{2}\lambda_k^T\lambda_k\right)^2 + b_{2k}\mu\sum_{i=1}^{n}\Theta(\lambda_{k,i})\left(\frac{1}{2}\lambda_{k,i}^2\right)^p\right] - \mu h_1\left(\frac{1}{2r}\tilde{\theta}^2\right)^p - \mu\left(4h_2-9\chi^{\frac{4}{3}}h_2\right)\left(\frac{1}{2r}\tilde{\theta}^2\right)^2 + D_3, \tag{75}$$

where $D_3 = D_2 + \frac{1}{2}\Gamma^T\Gamma$.

To address the switching function $\Theta(\varkappa_i)$ in (75), the following analysis is conducted:

(19) can be rewritten as

$$\Theta(\varkappa_i) = \begin{cases} 1, & |\varkappa_i| \ge \varepsilon, \\ 1-\cos^2\left(\frac{\varkappa_i\pi}{2\varepsilon}\right), & |\varkappa_i| < \varepsilon. \end{cases} \tag{76}$$

For $|\varkappa_i|<\varepsilon$, we have

$$\cos^2\left(\frac{\varkappa_i\pi}{2\varepsilon}\right)\left(\frac{1}{2}\varkappa_i^2\right)^p < \frac{1}{2^p}\varepsilon^{2p}. \tag{77}$$

Consequently, using $0<\mu\le\bar{\mu}$, one obtains

$$-\mu\Theta(\varkappa_i)\left(\frac{1}{2}\varkappa_i^2\right)^p = -\mu\left[1-\cos^2\left(\frac{\varkappa_i\pi}{2\varepsilon}\right)\right]\left(\frac{1}{2}\varkappa_i^2\right)^p < -\mu\left(\frac{1}{2}\varkappa_i^2\right)^p + \bar{\mu}\frac{\varepsilon^{2p}}{2^p}. \tag{78}$$

Applying this inequality to $v_{k,i}$ and $\lambda_{k,i}$, $k=1,2$, $i=1,\ldots,n$, one has

$$\dot{V} < -\sum_{k=1}^{2}\left[a_{2k-1}\mu\left(\frac{1}{2}v_k^Tv_k\right)^2 + a_{2k}\mu\sum_{i=1}^{n}\left(\frac{1}{2}v_{k,i}^2\right)^p + b_{2k-1}\mu\left(\frac{1}{2}\lambda_k^T\lambda_k\right)^2 + b_{2k}\mu\sum_{i=1}^{n}\left(\frac{1}{2}\lambda_{k,i}^2\right)^p\right] - \mu h_1\left(\frac{1}{2r}\tilde{\theta}^2\right)^p - \mu\left(4h_2-9\chi^{\frac{4}{3}}h_2\right)\left(\frac{1}{2r}\tilde{\theta}^2\right)^2 + D, \tag{79}$$

where $D = D_3 + \frac{n\bar{\mu}(a_2+a_4+b_2+b_4)\varepsilon^{2p}}{2^p}$.

By Lemma 2, we have

$$\dot{V} \le -\sum_{k=1}^{2}\left[a_{2k-1}\mu\left(\frac{1}{2}v_k^Tv_k\right)^2 + a_{2k}\mu\left(\frac{1}{2}v_k^Tv_k\right)^p + b_{2k-1}\mu\left(\frac{1}{2}\lambda_k^T\lambda_k\right)^2 + b_{2k}\mu\left(\frac{1}{2}\lambda_k^T\lambda_k\right)^p\right] - \mu h_1\left(\frac{1}{2r}\tilde{\theta}^2\right)^p - \mu\left(4h_2-9\chi^{\frac{4}{3}}h_2\right)\left(\frac{1}{2r}\tilde{\theta}^2\right)^2 + D \le -A\mu V^2 - B\mu V^p + D, \tag{80}$$

where $A = \frac{1}{5}\min\left\{a_1, \frac{a_3}{M^2}, b_1, b_3, 4h_2-9\chi^{4/3}h_2\right\}$ and $B = \min\left\{a_2, \frac{a_4}{M^p}, b_2, b_4, h_1\right\}$.

Then, according to Theorem 1, we can conclude that the closed-loop system is practical FxTS-PUBST.

2). By Theorem 1 and (80), we obtain

$$V \le \max\{V_0, V_T\}, \tag{81}$$

where V_0 denotes the initial value of the Lyapunov function V and

$$V_T = \min\left\{[D/(1-\lambda)B]^{\frac{1}{p}}, [D/(1-\lambda)A]^{\frac{1}{2}}\right\}. \tag{82}$$

From (81), it can be concluded that v_k, λ_k, $\tilde{\theta}$ are bounded. Further, by (35), we have that z_k and ϖ are bounded. Combining the boundedness of ϖ and (14), for the jth constraint interval $(t_{r,j}, t_{e,j})$ of e, one has

$$\rho_{d,i} < \delta_i(t) < \rho_{u,i}. \tag{83}$$

By (13), we have

$$\frac{\rho_{d,i}}{S} < e_i < \frac{\rho_{u,i}}{S}. \tag{84}$$

Combining (7), it can be concluded that:

- When $t\to\underline{t}_{r,j}^+$, $\frac{\rho_{d,i}}{S}=-\infty$ and $\frac{\rho_{u,i}}{S}=+\infty$.
- When $t=t_{r,j}$, $\frac{\rho_{d,i}}{S}=\rho_{d,i}$ and $\frac{\rho_{u,i}}{S}=\rho_{u,i}$.

This means that e_i gradually receives constraints during the transition time interval $(\underline{t}_{r,j}, t_{r,j}]$, and finally receives prescribed performance constraints $\rho_{u,i}$ and $\rho_{d,i}$ at $t\in[t_{r,j}, t_{e,j})$.

For $t\in[0,\underline{t}_{r,1}]\cup[t_{e,g},\underline{t}_{r,g+1}]\cup[t_{e,N}+\Delta T_2,\infty)$, there are additional transition time intervals $[t_{e,j}, t_{e,j}+\Delta T_2)$. In these intervals:

- (84) is satisfied.
- When $t=t_{e,j}$, $\frac{\rho_{d,i}}{S}=\rho_{d,i}$ and $\frac{\rho_{u,i}}{S}=\rho_{u,i}$.
- When $t\to t_{e,j}+\Delta T_2^-$, $\frac{\rho_{d,i}}{S}=-\infty$ and $\frac{\rho_{u,i}}{S}=+\infty$.

This indicates that e gradually loses its constraints during the transition time interval $[t_{e,j}, t_{e,j}+\Delta T_2)$. In addition, during the remaining time period, $\varpi=e$, which means that e is not constrained anymore.

The above analysis indicates that (ii) is valid.

3). According to (32), (35), $|\bar{\alpha}_2-\alpha_1|<\Gamma$ and the boundedness of v_2 and λ_2, we can conclude that $\dot{q}$ is bounded. According to the boundedness ϖ and Assumption 2, we have that q is bounded. In addition, according to the boundedness of ϖ, it can be inferred that ψ and ϕ are bounded. From the boundedness of $\tilde{\theta}$, we can conclude that $\hat{\theta}$ is bounded. Then, it is easy to obtain that α_1 and α_2 are bounded. Finally, by the event-triggered mechanism (58) and the boundedness of α_2, we have that τ is bounded. Therefore, the proof of (iii) is completed.

4). From the established event-triggered mechanism (58), it is known that $E_{k+1,i}(t)=\zeta_i(t)-\tau_i(t)$, $|E_{k+1,i}(t)|<\sigma_i|\tau_i|+\omega_i$, $E_{k+1,i}(t_{k,i})=0$, and $\lim_{t\to t_{k+1,i}}|E_{k+1,i}(t)|=\sigma_i|\tau_i|+\omega_i$. Let

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TABLE I
PARAMETERS OF THE ROBOTIC SYSTEM

Parameters	Descriptions	Values
m_1	Mass of joint 1	1.8 kg
m_2	Mass of joint 2	1.6 kg
J_1	Moment of joint 1	0.6 kg $\cdot$ m^2
J_2	Moment of joint 2	0.8 kg $\cdot$ m^2
L_1	Length of joint 1	1.7 m
L_2	Length of joint 2	1.5 m
$\bar{L}_1$	Mass center of joint 1	1.2 m
$\bar{L}_2$	Mass center of joint 2	0.8 m
g	Acceleration of gravity	9.8 m/s^2

$\Delta t = t_{k+1,i} - t_{k,i}$, we can get that $\lim_{\Delta t\to 0}|\frac{E_{k+1,i}(t_{k+1,i})-E_{k+1,i}(t_{k,i})}{\Delta t}| = |\dot{E}_i|$. On each interval $[t_{k,i}, t_{k+1,i})$, the applied torque τ_i is constant and $E_i = \zeta_i - \tau_i$, so $\dot{E}_i = \dot{\zeta}_i$ almost everywhere. Differentiating the continuous control signal in (58) and using $|\tanh x| \le 1$ and $\text{sech}^2 x \le 1$ yields $|\dot{E}_i| \le (1+\sigma_i)\left[\left(1+\frac{|v_{2,i}||\alpha_{2,i}|}{\iota_i}\right)|\dot{\alpha}_{2,i}| + \frac{|\alpha_{2,i}|^2+\omega_i^2}{\iota_i}|\dot{v}_{2,i}|\right]$. Since all signals are bounded and Remark 7, there exists $\varrho > 0$ such that $|\dot{E}_i| \le (1+\sigma_i)\left[\left(1+\frac{|v_{2,i}||\alpha_{2,i}|}{\iota_i}\right)|\dot{\alpha}_{2,i}| + \frac{|\alpha_{2,i}|^2+\omega_i^2}{\iota_i}|\dot{v}_{2,i}|\right] \le \varrho$. Furthermore, we obtain $\Delta t = t_{k+1,i} - t_{k,i} \ge (\sigma_i|\tau_i| + \omega_i)/\varrho \ge \omega_i/\varrho := F > 0$, where F is a positive constant independent of the triggering index k. Therefore, all inter-event intervals have the uniform constant lower bound F, which excludes Zeno behavior. □

Remark 9: The proposed control algorithm can be extended to handle constraint-free and deferred/infinite-time constraint cases without altering the controller structure. Specifically:

1) For the constraint-free case, $t_{r,1} = +\infty$.
2) For the deferred constraint case, $\underline{t}_{r,1} > 0$ and $t_{e,1} = +\infty$.
3) For the infinite-time constraint case, $t_{r,1} = 0$ and $t_{e,1} = +\infty$.

By modifying the stretch function based on the aforementioned analysis, the proposed control algorithm can be adapted to handle the aforementioned constraint cases.

V. SIMULATION EXAMPLE

A. Simulation Verification

In this paper, an example of a two-link rigid robotic system is provided to demonstrate the effectiveness of the proposed method. The dynamic model and parameters of the system are shown in (1) and Table I, respectively. Furthermore, other information of the system is displayed as follows [38]:

$$M(q) = \begin{bmatrix} p_1 + 2p_2\cos(q_2) & p_3 + p_2\cos(q_2) \\ p_3 + p_2\cos(q_2) & p_3 \end{bmatrix},$$
$$C(q,\dot{q}) = \begin{bmatrix} -p_2\sin(q_2)\dot{q}_2 & -p_2\sin(q_2)(\dot{q}_1+\dot{q}_2) \\ p_2\sin(q_2)\dot{q}_1 & 0 \end{bmatrix},$$
$$G(q) = \begin{bmatrix} p_4\cos(q_1) + p_5\cos(q_1+q_2) \\ p_6\cos(q_1+q_2) \end{bmatrix}, \quad (85)$$

where $p_1 = m_1\bar{L}_1^2 + m_2\bar{L}_2^2 + m_2L_1^2 + J_1 + J_2$, $p_2 = m_2L_1\bar{L}_2$, $p_3 = m_2\bar{L}_2^2 + J_2$, $p_4 = (m_1\bar{L}_2 + m_2L_1)g$, $p_5 = m_2L_2$, $p_6 = m_2L_2g$.

The initial values are selected as $q(0) = [q_1(0), q_2(0)]^T = [-0.5, 0.3]^T$, $\dot{q}(0) = [\dot{q}_1(0), \dot{q}_2(0)]^T = [0,0]^T$, $\bar{\alpha}_2(0) = [\bar{\alpha}_{2,1}(0), \bar{\alpha}_{2,2}(0)]^T = [3.039, -0.243]^T$, $\lambda_1(0) = [\lambda_{1,1}(0), \lambda_{1,2}(0)]^T = [0,0]^T$, $\lambda_2(0) = [\lambda_{2,1}(0), \lambda_{2,2}(0)]^T = [-3.039, 0.244]^T$, and $\hat{\theta}(0) = 0$. The reference trajectory is expressed as $q_d = [q_{d,1}, q_{d,2}] = [\sin(0.8t), \sin(0.8t)]^T$. The constrained time intervals are respectively chosen as $(t_{r,1}, t_{e,1}) = (3.5, 10)$ and $(t_{r,2}, t_{e,2}) = (12.5, 25)$. The design parameters are shown in Table II. The external disturbance is selected as

$$d(t) = \begin{cases} [10\sin t,\ 10\cos t]^T, & 5 < t \le 10, \\ [100,\ -100]^T, & 10.4 \le t < 11.2, \\ [100,\ -100]^T, & 16 \le t < 16.8, \\ [0,\ 0]^T, & \text{otherwise}. \end{cases} \quad (86)$$

The upper and lower constraint functions in (2) are selected as

$$\rho_{u,i}(t) = (\bar{\rho}_{u,i} - \underline{\rho}_{u,i})e^{-lt} + \underline{\rho}_{u,i},$$
$$\rho_{d,i}(t) = (\bar{\rho}_{d,i} - \underline{\rho}_{d,i})e^{-lt} + \underline{\rho}_{d,i}, \quad (87)$$

where $i = 1, 2$, $l = 1$, $\bar{\rho}_{u,1} = \bar{\rho}_{u,2} = 1.5$, $\bar{\rho}_{d,1} = \bar{\rho}_{d,2} = -1.5$, $\underline{\rho}_{u,1} = 0.075$, $\underline{\rho}_{u,2} = 0.1125$, $\underline{\rho}_{d,1} = -0.1125$, and $\underline{\rho}_{d,2} = -0.075$.

The simulation results are shown in Figs. 1–6. Fig. 1 depicts the curves of joint positions q_1, q_2 and reference trajectories $q_{d,1}$, $q_{d,2}$. Fig. 2 illustrates the curves of the tracking errors e_1 and e_2. As seen from Fig. 2, as the system approaches the prescribed settling-time upper bound $T = 2$, $|e_1|$ and $|e_2|$ gradually decrease and converge to a

TABLE II
THE DESIGN PARAMETERS

$a_1 = 4,\ a_2 = 4,\ a_3 = 1000,\ a_4 = 20,\ b_1 = b_2 = b_3 = b_4 = 4$
$\xi_2 = 0.05,\ r = 0.1,\ h_1 = 4,\ h_2 = 20,\ p = 0.55, \varepsilon = 0.7,\ \lambda = 0.9$
$\sigma_1 = \sigma_2 = 0.2,\ \omega_1 = \omega_2 = 3,\ \iota_1 = \iota_2 = 15$
$T = 2,\ T_s = 1.1,\ \Delta T_1 = 0.5,\ \Delta T_2 = 1$

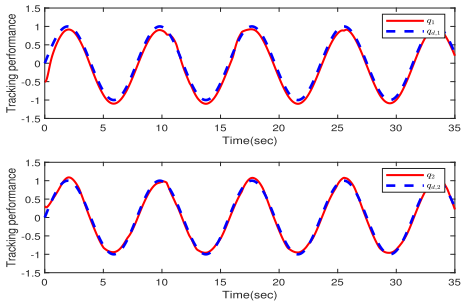


Fig. 1. Curves of q_1, q_2, $q_{d,1}$ and $q_{d,2}$.

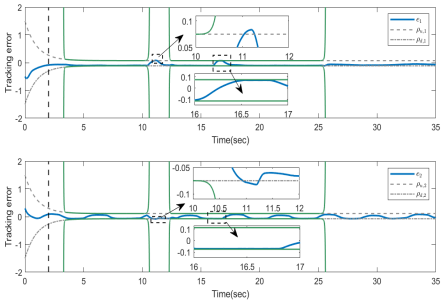


Fig. 2. Curves of tracking errors e_1 and e_2.

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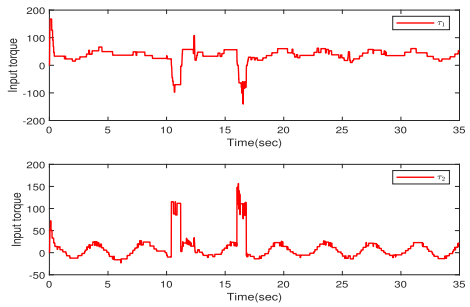


Fig. 3. Curves of input torques τ_1 and τ_2.

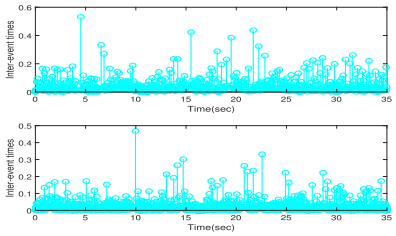


Fig. 4. Triggering time interval of the actual controllers τ_1 and τ_2.

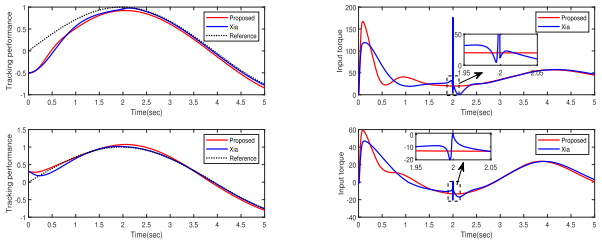


Fig. 5. Responses of states and controllers without measurement noise.

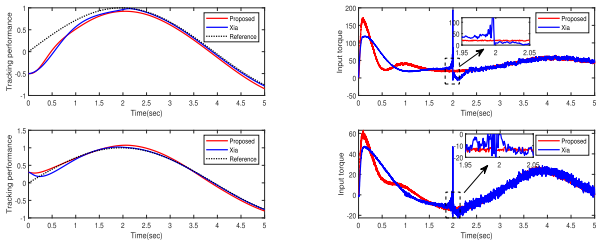


Fig. 6. Responses of states and controllers with measurement noise.

small neighborhood around zero. Additionally, Fig. 2 clearly demonstrates the dynamic error constraint process. During the interval $(\underline{t}_{r,1}, t_{r,1})$, the constraint trajectories are gradually stretched from infinity to the actual constraints $\rho_{u,i}$ and $\rho_{d,i}$, and maintain this state until time $t = t_{e,1}$. In the subsequent interval $(t_{e,1}, t_{e,1} + \Delta T_2)$, the constraint trajectories are re-stretched to infinity, thereby removing the constraint. In addition, identical large disturbance pulses are introduced over $[10.4, 11.2)$ and $[16, 16.8)$. The first pulse causes the tracking error to exceed the nominal upper bound $\rho_{u,i}$ during the constraint-release and subsequent unconstrained phases. In contrast, the prescribed constraints remain satisfied under the second pulse within the constrained interval $(12.5, 25)$. These results demonstrate the effectiveness of the proposed controller in handling intermittent constraints. Fig. 3 shows the input torques τ_1 and τ_2. Both disturbance pulses induce substantial torque variations. In particular, the pulse applied during the constrained phase requires a larger control effort to maintain the tracking errors within the prescribed bounds. Fig. 4 shows the triggering time intervals of these two controllers. Table III quantifies the update counts relative to periodic control with an update period of 0.01 s over the same 35 s simulation horizon. The event-triggered scheme requires 918 and 3011 updates for the two channels, respectively, compared with 3500 periodic updates per channel. The corresponding reductions are 73.77% and 13.97%, yielding an overall reduction of 43.87%.

TABLE III
COMPARISON OF CONTROL UPDATE COUNTS AND REDUCTION RATIOS

Channel	Periodic updates	Recorded triggers	Reduction (%)
Actuator 1	3500	918	73.77
Actuator 2	3500	3011	13.97
Total	7000	3929	43.87

B. Comparative Simulations

1) Comparison Under Measurement Noise: The PT control schemes for two-link rigid robotic systems [46] are compared to comparatively evaluate robustness against measurement noise. For the sake of fair comparison, the event-triggered mechanism and performance constraint are omitted in this example.

The system model is the same as that in the previous section. The state measurement is given by

$$\hat{x} = x + no(t), \tag{88}$$

where $n \in \{0, 1\}$, and $o(t)$ represents the measurement noise, which is generated by the MATLAB "Uniform Random Number" block with a range of $[-0.01, 0.01]$.

Based on [46], the PT controller without the event-triggered mechanism is implemented as

$$u = -2\eta_1 - \frac{1}{2}(\bar{m}\varrho + 1)\eta_2 - \mathrm{sgn}(\eta_2)\hat{d} - \hat{\theta}^T\phi(\omega),$$

where η_1 and η_2 are auxiliary error variables, $\hat{d}$ and $\hat{\theta}$ denote the adaptive estimates, and the remaining variables and adaptive laws follow [46]. The time-varying gain ϱ is defined as

$$\varrho = a + \frac{|\dot{\vartheta}|}{\vartheta}, \quad \vartheta(t) = \begin{cases} e^{\nu(T-t)} - 1, & 0 \le t < T, \\ e^{\nu(t-T)}, & t \ge T, \end{cases}$$

where $a > 0$, $\nu > 0$, and $T > 0$ is the prescribed settling time. Notably, ϱ becomes unbounded as $t \to T^-$. For the piecewise function $\vartheta(t)$, switching is performed 0.001 s before the prescribed time in the simulations to avoid numerical failure caused by the singular gain.

The controller parameters proposed in this paper remain unchanged. Figs. 5 and 6 depict the trajectories of state x and the corresponding control input signals for both controllers under noise-free and noisy conditions, respectively. Both the

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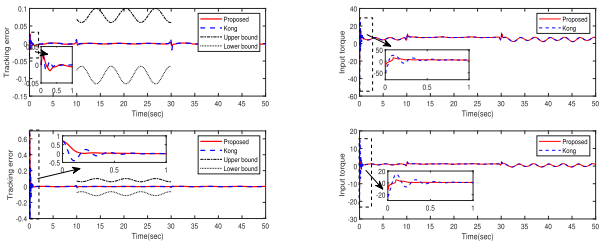


Fig. 7. Responses of states and controllers.

TABLE IV
QUANTITATIVE PERFORMANCE COMPARISON

Metric	Proposed method	Controller in [11]	Reduction (%)
Maximum $\lvert e_i \rvert$	[0.02790, 0.7000]	[0.02683, 0.7000]	[−4.00, 0.00]
RMSE	[0.002001, 0.02418]	[0.001987, 0.02349]	[−0.67, −2.93]
Peak $\lvert \tau_i \rvert$	[26.81, 11.24]	[51.44, 23.88]	[47.88, 52.93]
J_u	1927.69	2036.77	5.36

Paired values correspond to joints 1 and 2, respectively. All metrics are evaluated over 0–50 s, with $\mathrm{RMSE}_i = \sqrt{\frac{1}{50}\int_0^{50} e_i^2(t)\,\mathrm{d}t}$ and $J_u = \int_0^{50}\left(\tau_1^2(t)+\tau_2^2(t)\right)\,\mathrm{d}t$.

PT controller and the proposed controller successfully achieve state convergence within the PT. However, the control signal of the PT controller [46] exhibits a drastic surge as time t approaches the PT T. This anomaly primarily stems from the fact that the term ϱ tends to infinity as $t \to T$, and the early switching performed at $T - 0.001$ s. These effects significantly amplify the influence of measurement noise and make the control-signal chattering more pronounced. By contrast, the control signal of the proposed controller varies more smoothly.

2) Comparison With an Existing Intermittent-Constraint Controller: Existing intermittent-constraint control methods have been investigated for multi-input multi-output robotic systems. To evaluate the proposed method, comparative simulations are conducted with the controller in [11]. Since this controller is restricted to a single constrained interval, the constrained phase is selected as $(10, 30)$ s. For a fair comparison, both controllers are implemented without event triggering.

Both methods use the two-link manipulator model in [11]. To satisfy Property 2 required by the proposed method, $M_{11} = p_1 + 2p_2$ is replaced by $M_{11} = p_1 + 2p_2 \cos q_2$ for both methods. The disturbance, reference trajectory, and constraint boundaries follow the settings in [11]. The initial joint positions and velocities are identical.

The simulation results are presented in Fig. 7. The proposed method exhibits fewer oscillations in the tracking errors and control inputs during the initial transient. As summarized in Table IV, it reduces the peak torques of joints 1 and 2 by 47.88% and 52.93%, respectively, and decreases the integrated squared control input by 5.36%. Although the proposed method yields slightly higher RMSE values than the controller in [11], its stretch function requires the next constraint activation time to be available only $\Delta T_{\min}$ s before activation, rather than at the start of the unconstrained interval. This reduced requirement for advance timing information improves its practical applicability to intermittent-constraint control tasks.

VI. CONCLUSION

This paper conducts further research on practical fixed-time control for uncertain robotic systems with intermittent constraints. A new practical fixed-time stability theorem is proposed, and an adaptive controller is developed based on this theorem to enable the system to achieve practical FxTS-PUBST. Meanwhile, to address the problem of intermittent constraints, a novel stretch function is developed that does not require prior knowledge of the next constrained instant at the moment of switching to the unconstrained interval. However, this paper still has several limitations. Its effectiveness was only evaluated through simulations of a two-link robotic system, and actuator saturation, communication delays, and uncertain or rapidly changing constraint activation have not been explicitly considered. Future work will focus on experimental validation using higher-degree-of-freedom robotic systems and extending the proposed framework to address these practical issues.

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